

Chapter 5.4 - Cache Performance & Miss Penalty

Comprehensive Revision Flashcards • 95 Key Concepts & Practice Questions

#1	Q1: What are the two primary techniques explored for improving cache performance?
	Answer: Reducing the miss rate and reducing the miss penalty.
#2	Q2: CPU time can be divided into the clock cycles the CPU spends executing the program and the clock cycles it spends ____.
	Answer: waiting for the memory system
#3	Q3: What is the formula for CPU time in terms of execution cycles and memory-stall cycles?
	Answer: $\text{CPU time} = (\text{CPU execution clock cycles} + \text{Memory-stall clock cycles}) \times \text{Clock cycle time}.$
#4	Q4: Memory-stall clock cycles are primarily caused by what events?
	Answer: Cache misses.
#5	Q5: What is the formula for calculating Read-stall cycles?
	Answer: $\text{Read-stall cycles} = (\text{Reads/Program}) \times \text{Read miss rate} \times \text{Read miss penalty}.$
#6	Q6: In a write-through scheme, what are the two sources of write stalls?
	Answer: Write misses and write buffer stalls.
#7	Q7: Under what two conditions can write buffer stalls often be safely ignored?
	Answer: When the system has a reasonably deep write buffer (e.g., four+ words) and a memory that accepts writes much faster than the average write frequency.
#8	Q8: What simplified formula can be used for Memory-stall clock cycles if read and write penalties are the same and write buffer stalls are negligible?
	Answer: $\text{Memory-stall clock cycles} = (\text{Memory accesses/Program}) \times \text{Miss rate} \times \text{Miss penalty}.$
#9	Q9: What is the formula for memory-stall cycles per instruction?
	Answer: $\text{Memory-stall cycles / Instruction} = (\text{Misses / Instruction}) \times \text{Miss penalty}.$

#10	Q10: According to Amdahl's Law, what happens to the fraction of execution time spent on memory stalls if the processor is made faster but the memory system is not?
	Answer: The fraction of time spent on memory stalls will increase.
#11	Q11: What is a potential negative consequence of increasing cache size on performance?
	Answer: It can increase the hit time, which may increase the processor cycle time or add pipeline stages.
#12	Q12: What metric is used by designers to examine alternative cache designs by considering the time for both hits and misses?
	Answer: Average memory access time (AMAT).
#13	Q13: What is the formula for Average Memory Access Time (AMAT)?
	Answer: $AMAT = \text{Time for a hit} + (\text{Miss rate} \times \text{Miss penalty})$.
#14	Q14: In a _____ cache, a memory block can be placed in exactly one location.
	Answer: direct-mapped
#15	Q15: Term: Fully associative cache
	Answer: A cache structure in which a block can be placed in any location in the cache.
#16	Q16: How is a specific block found in a fully associative cache?
	Answer: All entries in the cache must be searched in parallel using comparators associated with each entry.
#17	Q17: Term: Set-associative cache
	Answer: A cache that has a fixed number of locations (at least two) where each block can be placed.
#18	Q18: Describe the two-step process of locating a memory block in a set-associative cache.
	Answer: A block is first directly mapped to a specific set, and then all blocks within that set are searched for a match.
#19	Q19: What formula determines the cache block for a memory block in a direct-mapped cache?
	Answer: $(\text{Block number}) \bmod (\text{Number of blocks in the cache})$.

#20	Q20: What formula determines the set for a memory block in a set-associative cache?
	Answer: (Block number) modulo (Number of sets in the cache).
#21	Q21: A direct-mapped cache can be considered a _____-way set-associative cache.
	Answer: one
#22	Q22: A fully associative cache with 'm' entries is equivalent to what kind of set-associative cache?
	Answer: An m-way set-associative cache with only one set.
#23	Q23: What is the primary advantage of increasing the degree of associativity in a cache?
	Answer: It usually decreases the miss rate.
#24	Q24: For a fixed cache size, what is the effect of doubling the associativity on the number of sets and the number of blocks per set?
	Answer: It halves the number of sets and doubles the number of blocks per set.
#25	Q25: How does increasing cache associativity by a factor of 2 affect the size of the address's index and tag fields?
	Answer: It decreases the size of the index by 1 bit and increases the size of the tag by 1 bit.
#26	Q26: How many index bits are used in a fully associative cache?
	Answer: Zero; there is no index, as the entire cache is searched.
#27	Q27: What hardware is required to implement a four-way set-associative cache that is not needed for a direct-mapped cache?
	Answer: Four parallel comparators and a 4-to-1 multiplexor.
#28	Q28: Term: Content Addressable Memory (CAM)
	Answer: A circuit that combines comparison and storage, returning the index of a matching data entry rather than requiring an address.
#29	Q29: For high degrees of associativity (e.g., eight-way and above), what type of hardware is often used instead of standard SRAMs and comparators?
	Answer: Content Addressable Memories (CAMs).

#30	Q30: In an associative cache, what is the most commonly used scheme for choosing which block to replace on a miss?
	Answer: Least Recently Used (LRU).
#31	Q31: Term: Least Recently Used (LRU)
	Answer: A replacement scheme in which the block replaced is the one that has been unused for the longest time.
#32	Q32: How does increasing associativity affect the total number of tag bits in a cache of a fixed size?
	Answer: It increases the total number of tag bits because each block's tag field becomes larger.
#33	Q33: Term: Multilevel cache
	Answer: A memory hierarchy with multiple levels of caches, rather than just a single cache and main memory.
#34	Q34: How does a second-level cache improve performance when a miss occurs in the primary cache?
	Answer: If the data is in the second-level cache, the miss penalty is its access time, which is much lower than accessing main memory.
#35	Q35: In a system with a primary cache and a secondary cache, what is the formula for Total CPI?
	Answer: $\text{Total CPI} = \text{Base CPI} + (\text{Primary stalls per instruction}) + (\text{Secondary stalls per instruction})$.
#36	Q36: In a multilevel cache design, the primary (L1) cache is often designed to minimize ____.
	Answer: hit time
#37	Q37: In a multilevel cache design, the secondary (L2) cache is often designed to minimize ____.
	Answer: miss rate
#38	Q38: How does a primary cache in a multilevel system typically differ from an optimal single-level cache in terms of size and block size?
	Answer: It is often smaller and may use a smaller block size.
#39	Q39: How does a secondary cache in a multilevel system typically differ from an optimal single-level cache in terms of size and associativity?
	Answer: It is much larger and often uses higher associativity.

#40	Q40: Why might an algorithm like Quicksort outperform Radix Sort in practice, even if Radix Sort executes fewer instructions for large arrays?
	Answer: Quicksort can have significantly fewer cache misses, making better use of the memory hierarchy.
#41	Q41: What is the goal of a blocked algorithm in the context of software optimization for caches?
	Answer: To maximize accesses to data loaded into the cache before it is replaced, thereby improving temporal locality.
#42	Q42: The software optimization technique of operating on submatrices instead of entire rows or columns is known as ____.
	Answer: blocking
#43	Q43: In matrix multiplication, blocking improves performance by exploiting a combination of spatial locality for one matrix and ____ locality for another.
	Answer: temporal
#44	Q44: Term: Global miss rate
	Answer: The fraction of references that miss in all levels of a multilevel cache.
#45	Q45: Term: Local miss rate
	Answer: The fraction of references to one specific level of a cache that miss at that level.
#46	Q46: Why is the local miss rate of a secondary cache typically much higher than the global miss rate?
	Answer: Because the primary cache filters out most accesses with good spatial and temporal locality, leaving only harder-to-hit references for the secondary cache.
#47	Q47: For out-of-order processors, performance evaluation requires simulation because the processor can hide miss latency by ____.
	Answer: finding other work to do (executing other instructions)
#48	Q48: Term: Autotuning
	Answer: An approach where numerical libraries parameterize algorithms and search at runtime to find the best parameter combination for a specific computer's memory hierarchy.

#49	Q49: Which of the following is generally true about a design with multiple levels of caches?
	Answer: First-level caches are more concerned about hit time, and second-level caches are more concerned about miss rate.
#50	Q50: Which cache designer guideline regarding memory bandwidth and block size is generally valid?
	Answer: The higher the memory bandwidth, the larger the cache block can be without a significant performance penalty.
#51	Q51: What are the two primary techniques mentioned for improving cache performance?
	Answer: Reducing the miss rate and reducing the miss penalty.
#52	Q52: How can the miss rate of a cache be reduced?
	Answer: By reducing the probability that two distinct memory blocks will contend for the same cache location, for example by using associativity.
#53	Q53: What technique is used to reduce the miss penalty in a cache hierarchy?
	Answer: Adding an additional level to the hierarchy, known as multilevel caching.
#54	Q54: The memory-stall clock cycles in a CPU primarily come from ____.
	Answer: cache misses
#55	Q55: For a write-through cache scheme, what are the two main sources of stalls?
	Answer: Write misses and write buffer stalls.
#56	Q56: When can write buffer stalls often be safely ignored in performance calculations?
	Answer: When the system has a reasonably deep write buffer (e.g., four or more words) and memory can accept writes faster than the average write frequency.
#57	Q57: What is the simplified formula for memory-stall cycles, assuming negligible write buffer stalls and equal read/write penalties?
	Answer: Memory-stall clock cycles = (Memory accesses/Program) × Miss rate × Miss penalty.
#58	Q58: How can memory-stall cycles be expressed on a per-instruction basis?
	Answer: Memory-stall clock cycles = (Instructions/Program) × (Misses/Instruction) × Miss penalty.

#59	Q59: According to Amdahl's Law, what happens to the fraction of execution time spent on memory stalls if a processor's CPI is reduced but the memory system is not improved?
	Answer: The fraction of time spent on memory stalls will increase.
#60	Q60: Term: Average Memory Access Time (AMAT)
	Answer: Definition: The average time to access memory, considering both hits and misses, calculated as: $AMAT = \text{Time for a hit} + (\text{Miss rate} \times \text{Miss penalty})$.
#61	Q61: What hardware components make fully associative placement impractical for large caches?
	Answer: A large number of parallel comparators, one associated with each cache entry.
#62	Q62: How is the location for a memory block determined in a set-associative cache?
	Answer: A block is directly mapped to a specific set, and then it can be placed in any of the elements within that set.
#63	Q63: In a direct-mapped cache, how is the cache block for a memory block determined?
	Answer: It is determined by the formula: $(\text{Block number}) \bmod (\text{Number of blocks in the cache})$.
#64	Q64: In a set-associative cache, how is the set for a memory block determined?
	Answer: It is determined by the formula: $(\text{Block number}) \bmod (\text{Number of sets in the cache})$.
#65	Q65: An m-block fully associative cache can be considered a ____-way set-associative cache.
	Answer: m
#66	Q66: What is the main disadvantage of increasing the degree of associativity in a cache?
	Answer: A potential increase in the hit time due to more complex hardware (comparators and multiplexors).
#67	Q67: How are all the tags within a selected set of a set-associative cache searched to ensure a fast hit time?
	Answer: They are searched in parallel.
#68	Q68: What are the three portions of a memory address used by a set-associative or direct-mapped cache?
	Answer: The tag, the index, and the block offset.

#69	Q69: For a fixed cache size, what is the effect of doubling the associativity on the number of sets and the size of the index field?
	Answer: It halves the number of sets and decreases the size of the index field by 1 bit.
#70	Q70: For a fixed cache size, what is the effect of doubling the associativity on the size of the tag field?
	Answer: It increases the size of the tag field by 1 bit.
#71	Q71: In modern cache design, what level of associativity is typically built using standard SRAMs and comparators versus using CAMs?
	Answer: Two-way and four-way associativity are often built with SRAMs and comparators, while eight-way and higher associativity are often built using CAMs.
#72	Q72: What is the most commonly used block replacement scheme for associative caches?
	Answer: Least Recently Used (LRU).
#73	Q73: As cache associativity increases, what happens to the difficulty of implementing a true LRU replacement policy?
	Answer: It becomes harder to implement.
#74	Q74: What is the purpose of a second-level (L2) cache in a memory hierarchy?
	Answer: It is accessed on a primary (L1) cache miss to reduce the miss penalty by providing faster access than main memory.
#75	Q75: What is the formula for the total CPI in a system with two levels of caching?
	Answer: $\text{Total CPI} = \text{Base CPI} + (\text{Primary misses/instruction} \times \text{L2 hit penalty}) + (\text{Global misses/instruction} \times \text{Main memory penalty})$.
#76	Q76: In a multilevel cache system, what is the primary design focus for the first-level (L1) cache?
	Answer: Minimizing hit time to enable a shorter clock cycle or fewer pipeline stages.
#77	Q77: In a multilevel cache system, what is the primary design focus for the second-level (L2) cache?
	Answer: Minimizing the miss rate to reduce the penalty of long main memory access times.

#78	Q78: How do the size and block size of a primary cache in a multilevel system typically compare to an optimal single-level cache?
	Answer: The primary cache is often smaller and may use a smaller block size.
#79	Q79: How do the size and associativity of a secondary cache in a multilevel system typically compare to an optimal single-level cache?
	Answer: The secondary cache is much larger and often uses higher associativity.
#80	Q80: Despite requiring more instructions, why can Quicksort outperform Radix Sort in terms of actual execution time on modern machines?
	Answer: Quicksort often has significantly fewer cache misses, making better use of the memory hierarchy.
#81	Q81: What is the primary goal of software optimization via 'blocking'?
	Answer: To maximize accesses to data loaded into the cache before it's replaced, thereby improving temporal locality and reducing cache misses.
#82	Q82: Instead of operating on entire rows or columns of an array, blocked algorithms operate on ____.
	Answer: submatrices or blocks
#83	Q83: In the context of blocked matrix multiplication, the size of the submatrix is determined by a parameter called the ____.
	Answer: blocking factor
#84	Q84: Blocking in matrix multiplication improves performance by exploiting a combination of ____ and ____ locality.
	Answer: spatial; temporal
#85	Q85: What is the approximate improvement in the number of memory words accessed when using blocking for matrix multiplication?
	Answer: The number of memory words accessed is reduced by a factor roughly equal to the BLOCKSIZE.
#86	Q86: Besides reducing cache misses, what other optimization can blocking be used for?
	Answer: It can help with register allocation by using a block size small enough to be held in registers, minimizing loads and stores.

#87	Q87: Why is the local miss rate of a secondary cache typically much higher than its global miss rate?
	Answer: Because the primary cache filters out accesses with good spatial and temporal locality, leaving only more difficult references for the secondary cache.
#88	Q88: How is the memory stall calculation for out-of-order processors different from simpler models?
	Answer: It must account for the 'overlapped miss latency,' which is the portion of the miss penalty that the processor can hide by executing other instructions.
#89	Q89: As a general guideline, an out-of-order processor can often hide the miss penalty for a miss that hits in the _____ cache, but rarely for a miss to the _____ cache.
	Answer: L2; L1
#90	Q90: Which statement is generally true about designing a multilevel cache?
	Answer: First-level caches are more concerned about hit time, and second-level caches are more concerned about miss rate.
#91	Q91: How does higher memory bandwidth generally influence the optimal cache block size?
	Answer: Higher memory bandwidth supports the use of a larger cache block size, as the penalty for fetching a larger block is reduced.
#92	Q92: For a processor with a 1 ns clock, a miss penalty of 20 cycles, a miss rate of 0.05 misses/instruction, and a hit time of 1 cycle, what is the AMAT?
	Answer: $AMAT = 1 + (0.05 * 20) = 2$ clock cycles, or 2 ns.
#93	Q93: A processor has a CPI of 2 without memory stalls, a 100-cycle miss penalty, an instruction cache miss rate of 2%, and a data cache miss rate of 4%. What are the memory-stall cycles per instruction if 36% of instructions are loads/stores?
	Answer: Instruction stalls: $1 * 0.02 * 100 = 2.0$. Data stalls: $0.36 * 0.04 * 100 = 1.44$. Total stall cycles per instruction = $2.0 + 1.44 = 3.44$.
#94	Q94: Consider a cache of 4096 blocks, a 4-word block size, and a 64-bit address. For a two-way set-associative design, how many sets are there?
	Answer: There are $4096 / 2 = 2048$ sets.

#95	Q95: Consider a cache of 4096 blocks, a 4-word block size, and a 64-bit address. For a two-way set-associative design, how many bits are in the tag?
	Answer: The address has 60 bits for tag+index. The index needs $\log_2(2048) = 11$ bits. So, the tag is $60 - 11 = 49$ bits.